

CANONICAL LESSON

Resistance Signals Require Stewardship, Not Panic

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C27
Notebook	V7_B_N07
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Interpret antimicrobial-resistance signals with laboratory quality, sampling context, clinical relevance, uncertainty, and stewardship authority.

Learning outcomes

Why this matters

Decontextualized resistance signals can cause panic, inappropriate prescribing changes, or false reassurance.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

Worked cross-sector example

A resistance increase is reviewed with specimen mix, testing coverage, facility context, and prescribing practice.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Deduplicate isolates.
- 2.

- Handle breakpoint changes.
- 3.
- Explain why resistance rates are not prescribing rules.

Answer key

- 1.
- Use the agreed first-isolate-per-patient/episode/specimen rule.
- 2.
- Reinterpret stored measurements where valid or stratify by breakpoint version; do not splice incomparable rates.
- 3.
- Treatment also depends on syndrome, specimen, patient factors, severity, local guidance, toxicity, availability, and clinician judgment.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.